In [1]:
```python
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

In [4]:
```python
# Load Dataset into DataFrame
from sklearn.datasets import load_iris
df = pd.read_csv('/content/iris.csv')
```

In [5]:
```python
# Scan Dataset
print(df.head())
```

```
   sepal_length  sepal_width  petal_length  petal_width species
0           5.1          3.5           1.4          0.2  setosa
1           4.9          3.0           1.4          0.2  setosa
2           4.7          3.2           1.3          0.2  setosa
3           4.6          3.1           1.5          0.2  setosa
4           5.0          3.6           1.4          0.2  setosa
```

In [6]:
```python
print(df.info())
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
 4   species       150 non-null    object 
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
None
```

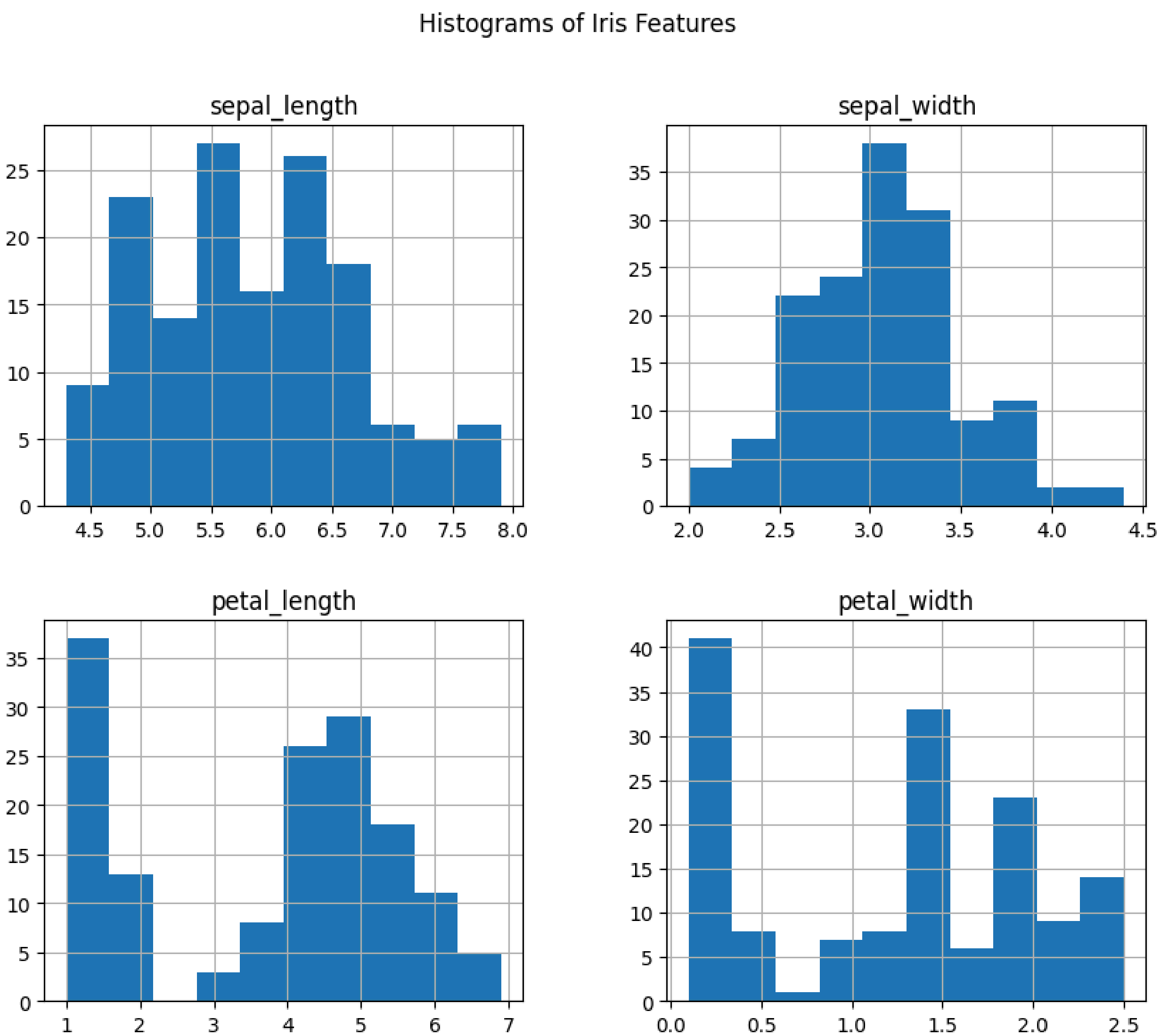
In [7]:
```python
print(df.describe())
```

```
       sepal_length  sepal_width  petal_length  petal_width
count    150.000000   150.000000    150.000000   150.000000
mean       5.843333     3.054000      3.758667     1.198667
std        0.828066     0.433594      1.764420     0.763161
min        4.300000     2.000000      1.000000     0.100000
25%        5.100000     2.800000      1.600000     0.300000
50%        5.800000     3.000000      4.350000     1.300000
75%        6.400000     3.300000      5.100000     1.800000
max        7.900000     4.400000      6.900000     2.500000
```

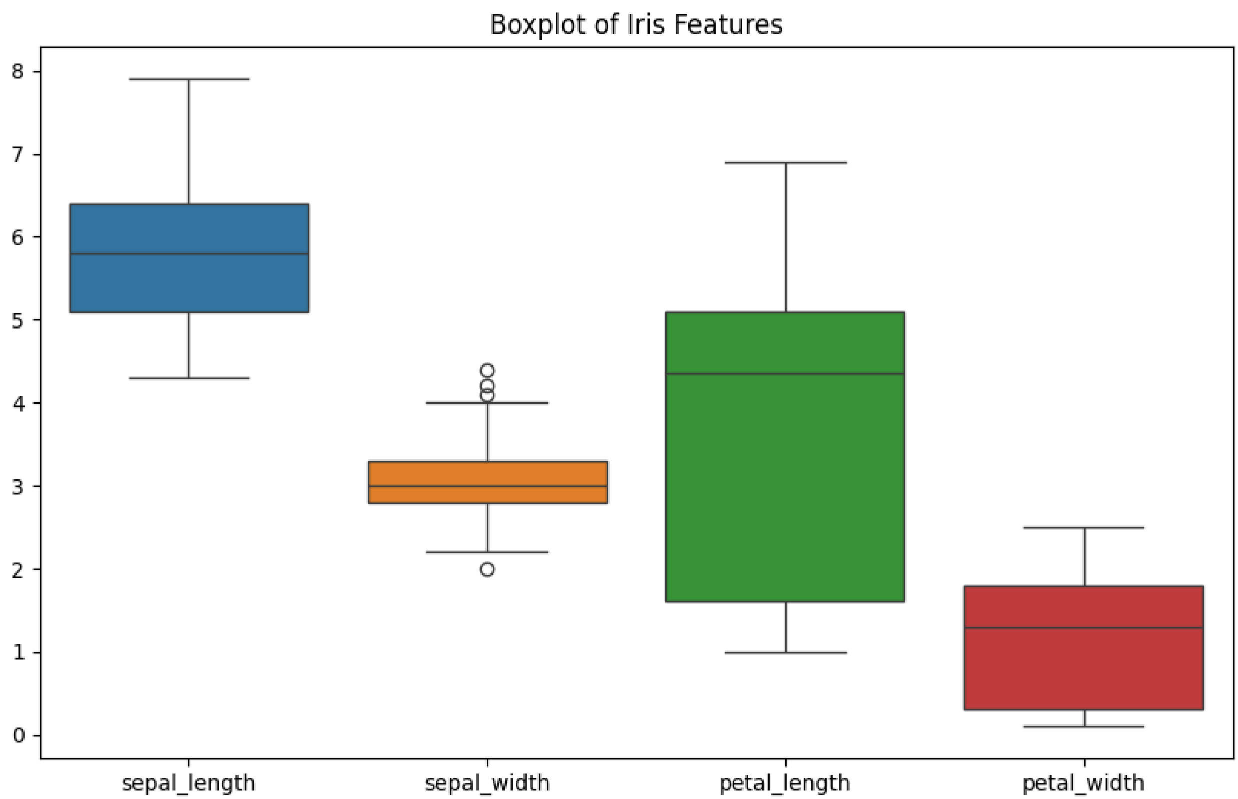
In [8]:
```python
# 1. Features and Their Types
print(df.dtypes)
```

```
sepal_length    float64
sepal_width     float64
petal_length    float64
petal_width     float64
species          object
dtype: object
```

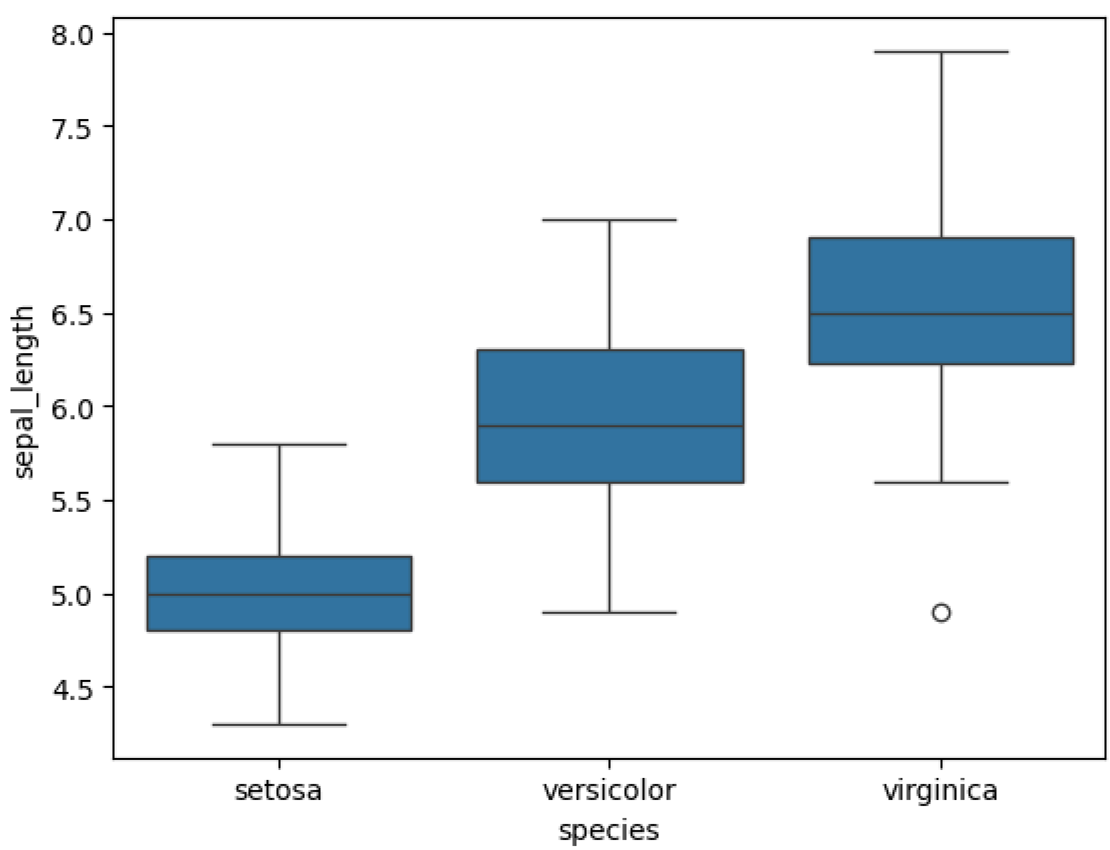
In [9]:
```python
# 2. Histogram for Each Feature
df.hist(figsize=(10,8))
plt.suptitle("Histograms of Iris Features")
plt.show()
```



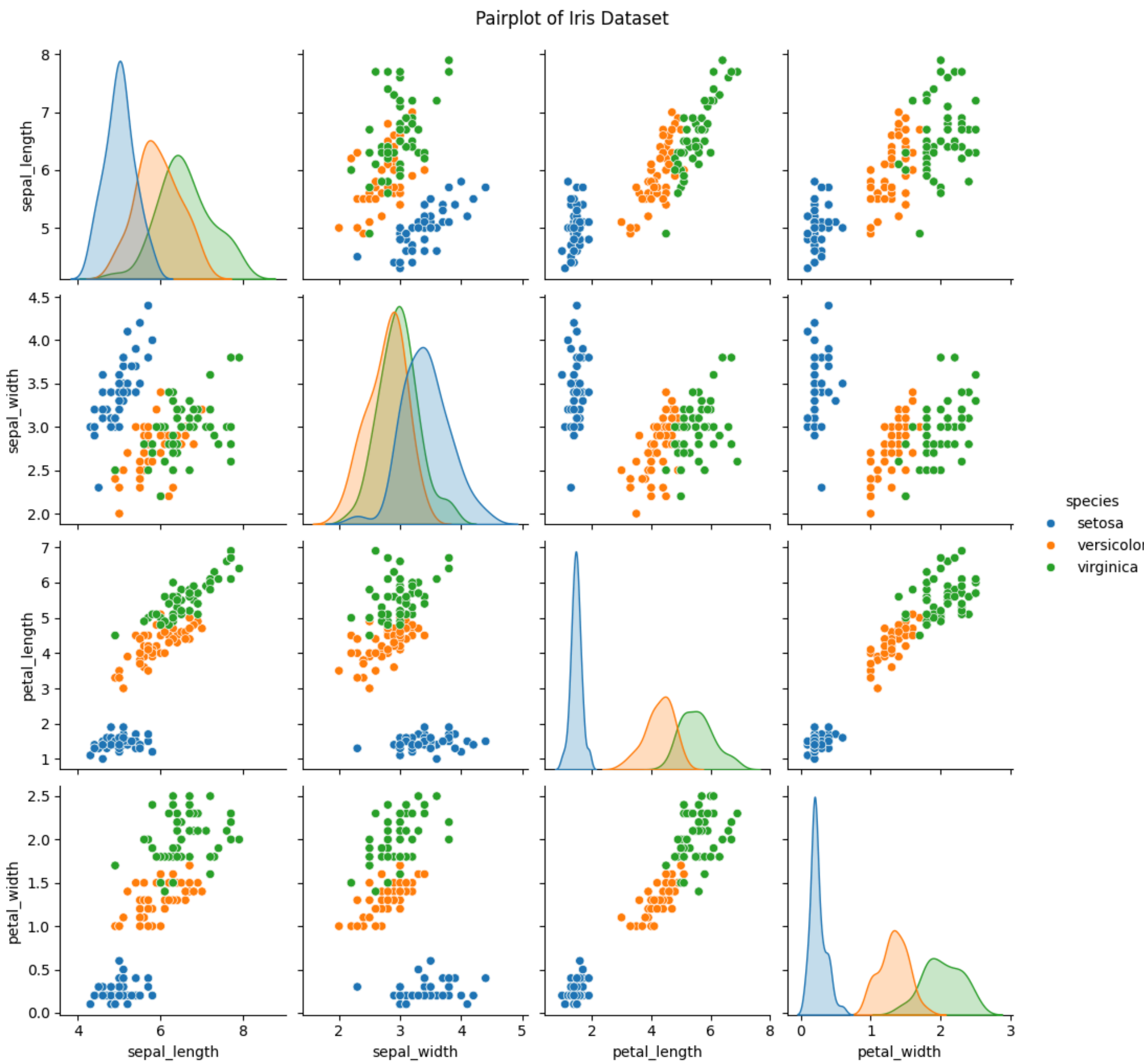
In [10]:
```python
# 3. Boxplot for Each Feature
plt.figure(figsize=(10,6))
sns.boxplot(data=df)
plt.title("Boxplot of Iris Features")
plt.show()
```



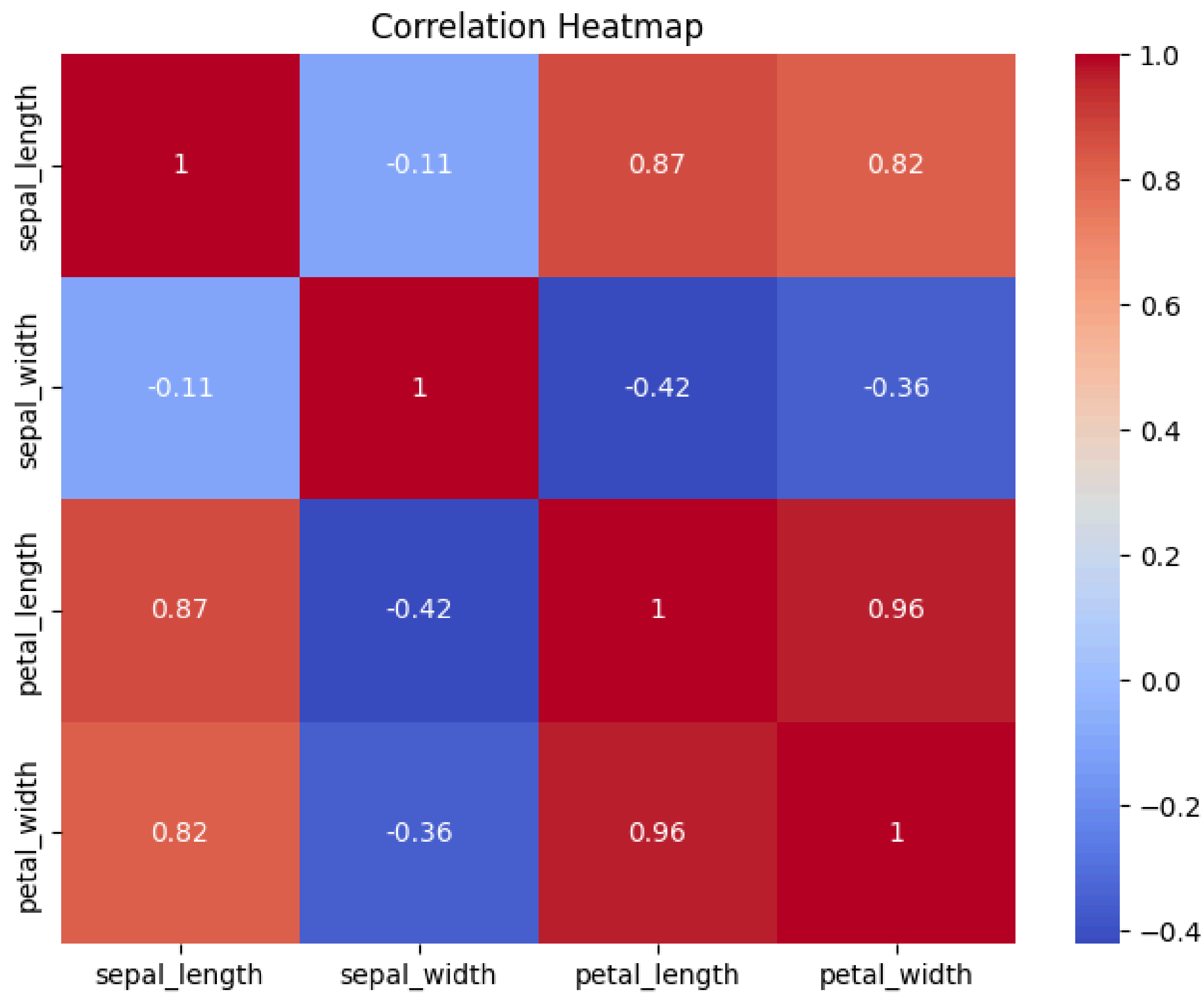
In [12]:
```python
# 4. Compare Distributions & Identify Outliers
sns.boxplot(x='species', y='sepal_length', data=df)
plt.show()
```



In [13]:
```
# 5. Visualizations for Iris Dataset
# 5: Pairplot
sns.pairplot(df, hue='species')
plt.suptitle("Pairplot of Iris Dataset", y=1.02)
plt.show()
```

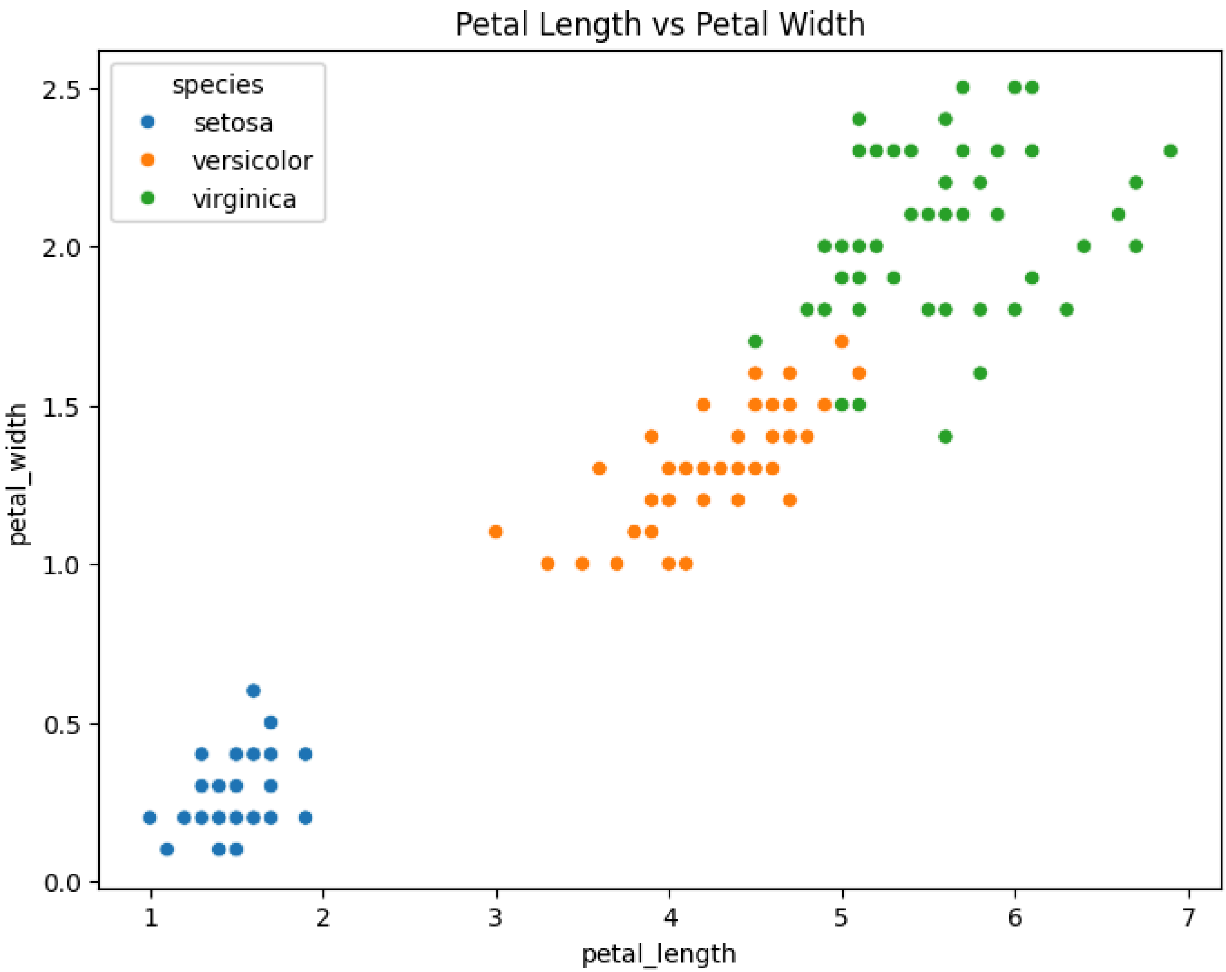


In [14]:
```
# 6.Heatmap
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

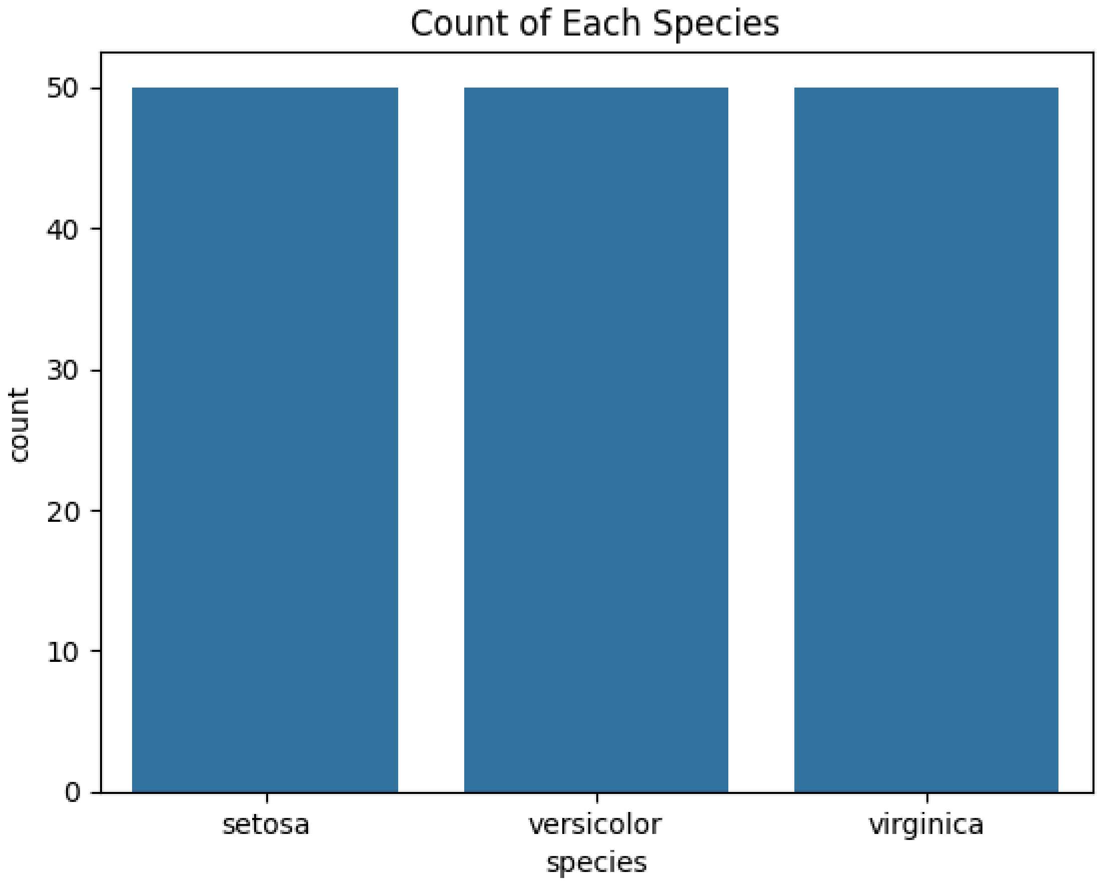


In [16]:

```
# 7: Scatter Plot
plt.figure(figsize=(8,6))
sns.scatterplot(x='petal_length', y='petal_width', hue='species', data=df)
plt.title("Petal Length vs Petal Width")
plt.show()
```



In [17]:
```
# 9: Count Plot (Categorical Distribution)
sns.countplot(x='species', data=df)
plt.title("Count of Each Species")
plt.show()
```


Count of Each Species
count
0
10
20
30
40
50
setosa
versicolor
virginica
species

In [ ]: